

Get Set Up: PostgreSQL + Bluebox

Docker, psql, and the Sample Database

Setup Guide — Beginning Postgres Workshop

PGSummit NYC

Training Materials

All slides, exercises, and Docker setup:

github.com/ryanbooz/postgres-full-day-training

Sample database:

github.com/ryanbooz/bluebox

Tip: on a Mac, Postgres.app includes a working psql client — no separate install needed.

Let's Get Connected!

Setup Overview

1. Clone the training repo
2. Start PostgreSQL (Docker)
3. Install a psql client
4. Connect to PostgreSQL
5. Create the database and load Bluebox
6. Verify everything works

~10-15 minutes

Prerequisites

You need **Docker Desktop** installed and running

- **Mac/Windows:** docker.com/products/docker-desktop
- **Linux:** Docker Engine + Docker Compose

Verify it's working:

```
docker --version
```

```
docker compose version
```

Step 1: Clone the Repository

```
## Clone the training repo
```

```
git clone https://github.com/ryanbooz/postgres-full-day-training.git
```

```
## Navigate into the folder
```

```
cd postgres-full-day-training
```

Or download as ZIP from GitHub if you don't have git.

Step 2: Start PostgreSQL

```
## Start PostgreSQL container  
docker compose up -d
```

```
## Verify it's running  
docker ps
```

You should see postgres-training running on port 5432.

CONTAINER ID	IMAGE	STATUS	PORTS
abc123...	postgis/postgis:18-3.6	Up 10 seconds	0.0.0.0:5432->5432/tcp

Step 3: Install a psql Client

You need a way to connect to PostgreSQL. Choose one:

Mac

- `brew install libpq (client only)`
- `brew install postgresql@18`
- Postgres.app (includes psql)

Linux

- `apt install postgresql-client`
- `yum install postgresql`

Windows

- PostgreSQL installer ([postgresql.org](https://www.postgresql.org))
- pgAdmin (standalone)

Cross-Platform GUIs

- pgAdmin
- DBeaver
- TablePlus
- Azure Data Studio
- Beekeeper Studio

Mac: Homebrew (Recommended)

```
## Install just the client tools (no server)
```

```
brew install libpq
```

```
## Add to your PATH
```

```
echo 'export PATH="/opt/homebrew/opt/libpq/bin:$PATH"' >> ~/.zshrc
```

```
source ~/.zshrc
```

```
## Verify
```

```
psql --version
```

Alternative: Install Postgres.app and use its bundled psql.

Windows / GUI Users

pgAdmin is a great option:

1. Download from pgadmin.org
2. Add a new server connection:
 - Host: localhost
 - Port: 5432
 - Username: postgres
 - Password: training

pgAdmin includes a Query Tool that works like psql.

Step 4: Connect to PostgreSQL

```
## Connect using psql
```

```
psql postgresql://postgres:training@localhost:5432/postgres
```

You should see:

```
psql (18.x)
```

```
Type "help" for help.
```

```
postgres=#
```

You're connected! 🎉

Step 5: Create Database and Load Bluebox

From psql, create the database:

```
CREATE DATABASE bluebox;  
\c bluebox  
CREATE EXTENSION postgis;
```

Then exit psql (\q) and load the data:

Download Bluebox files

```
curl -LO https://raw.githubusercontent.com/ryanbooz/bluebox/main/bluebox_schema.sql  
curl -LO https://raw.githubusercontent.com/ryanbooz/bluebox/main/bluebox_data.sql.gz  
gunzip bluebox_data.sql.gz
```

Load schema and data

```
psql postgresql://postgres:training@localhost:5432/bluebox -f bluebox_schema.sql  
psql postgresql://postgres:training@localhost:5432/bluebox -f bluebox_data.sql
```

Step 6: Verify Your Setup

Connect to the bluebox database:

```
psql postgresql://postgres:training@localhost:5432/bluebox
```

Run a test query:

```
SELECT COUNT(*) FROM bluebox.film;
```

```
count
```

```
-----
```

```
7836
```

If you see 7836 films, you're all set! 

Troubleshooting: Last Resort

If you can't install psql locally, use Docker:

```
## Connect via docker exec
```

```
docker exec -it postgres-training psql -U postgres -d bluebox
```

This works but isn't ideal for learning psql workflows.

Common issues:

- Docker not running → Start Docker Desktop
- Port conflict → Check if another Postgres is using 5432
- Permission denied → Run Docker Desktop as admin (Windows)

Loading Data Without Local psql or a GUI Tool

Bluebox's schema and data files are meant to be run with `psql -f` (they use COPY, not portable INSERTs) — most GUI query tools (pgAdmin, DBeaver, etc.) can't execute them directly.

If that happens, load the files using the psql that's already running inside the container.

1. Copy the files into the container

```
docker cp bluebox_schema.sql postgres-training:/tmp/  
docker cp bluebox_data.sql postgres-training:/tmp/
```

2. Load them from inside the container

```
docker exec -it postgres-training \  
psql -U postgres -d bluebox -f /tmp/bluebox_schema.sql  
  
docker exec -it postgres-training \  
psql -U postgres -d bluebox -f /tmp/bluebox_data.sql
```

You're Ready!

Keep this environment around — it stays useful for all three workshop sessions, and afterward for you to keep exploring on your own.

See you at the workshop!